

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

x	-11	-10	-9	-8
f(x)	21	18	15	12

The table above shows some values of x and their corresponding values $f(x)$ for the linear function f . What is the x -intercept of the graph of $y = f(x)$ in the xy -plane?

- Answer
- A. $(-3,0)$
- B. $(-4,0)$
- C. $(-9,0)$
- D. $(-12,0)$

Correct Answer: B

Rationale

Choice B is correct. The equation of a linear function can be written in the form $y = mx + b$, where $y = f(x)$, m is the slope of the graph of $y = f(x)$, and b is the y -coordinate of the y -intercept of the graph. The value of m can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$.

According to the table, the points $(-11,21)$ and $(-10,18)$ lie on the graph of $y = f(x)$. Using these two points in the slope formula yields $m = \frac{18 - 21}{-10 + 11}$, or -3 . Substituting -3 for m in the slope-intercept form of the equation yields $y = -3x + b$. The value of b can be found by substituting values from the table and solving; for example, substituting the coordinates of the point $(-11,21)$ into the equation $y = -3x + b$ gives $21 = -3(-11) + b$, which yields $b = -12$. This means the function given by the table can be represented by the equation $y = -3x - 12$. The value of the x -intercept of the graph of $y = f(x)$ can be determined by finding the value of x when $y = 0$. Substituting $y = 0$ into $y = -3x - 12$ yields $0 = -3x - 12$, or $x = -4$. This corresponds to the point $(-4,0)$.

Choice A is incorrect and may result from substituting the value of the slope for the x -coordinate of the x -intercept. Choice C is incorrect and may result from a calculation error. Choice D is incorrect and may result from substituting the y -coordinate of the y -intercept for the x -coordinate of the x -intercept.